

# Parth Brahmhatt

Department of Chemical and Biological Engineering, University of Wisconsin-Madison  
Email: [parthb1499@gmail.com](mailto:parthb1499@gmail.com) | Cell: [+1 608 217 6067](tel:+16082176067) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

## EDUCATION

**University of Wisconsin-Madison, USA** 2023–2027  
Ph.D., Chemical and Biological Engineering (*Process Systems Engineering*) GPA: 3.84/4.00  
M.S., Industrial and Systems Engineering (*Operations Research, Optimization, and Analytics*)  
Mathematical Optimization, Machine Learning, Controls, Stochastic programming.

**Indian Institute of Technology (IIT)- Bombay, India** 2021–2023  
M.Tech., Chemical Engineering CGPA: 9.76/10.00  
Digital Twins, Fault detection and diagnosis, Optimization, Deep Learning, Data Science.

**The Maharaja Sayajirao University of Baroda (MSU), India** 2016–2020  
B.Eng., Chemical Engineering GPA: 3.98/4.00 [*Gold Medal*]

## SKILLS

- Optimization: Julia- JuMP.jl; Python- Scipy, Gurobipy, Pyomo, CVXPY; CPLEX.
- ML and data Science: Tensorflow, Keras, Pytorch, Pytorch Geometric Temporal, NetworkX.
- Generative AI: LangChain, LangGraph, AI Agent Systems, RAG, Prompt Engineering.
- Software: Aspen Plus, EPANET, MS Office, R, MATLAB & Simulink,  $\text{\LaTeX}$ .

## EXPERIENCE

**University of Wisconsin- Madison** 2023–Present  
Graduate Research Assistant, Circular Economy Systems Engineering Group  
*PI- Prof. Styliana Avraamidou, UW- Madison*

- Developing stochastic optimization models for the integrated planning and scheduling of modular Power-to-X plants, enabling optimal strategies for participation in uncertain electricity markets.
- Designed a Benders decomposition framework leveraging multi-parametric programming to significantly improve the computational efficiency of solving large-scale stochastic problems.
- Developed a framework for explicit Model Predictive Control (MPC) by applying multi-parametric programming to represent and solve control laws derived from ReLU neural networks.
- Working on A Multi-scale Optimization of The Water-energy-carbon Nexus to assess the impact of Data Centers on urban energy infrastructure planning.

**Argonne National Laboratory** May 2025-Aug 2025  
Givens Associate, Mathematics and Computer Science Division

- Formulated non-linear optimization models to solve inverse problems in tomographic reconstruction, specifically incorporating and correcting for signal drift.
- Designed a multi-grid optimization framework using proximal algorithms to efficiently solve high-dimensional tomography reconstruction problems.

**Indian Institute of Technology (IIT)- Bombay, India** 2022–2023  
Graduate Researcher, SCEPTRE Lab- PSE Group  
*PIs- Prof. Ravi Gudi (IIT- Bombay) & Prof. Abhilasha Maheshwari (IIT- Jodhpur)*

- *Thesis:* Digital Twin Assisted Decision Support System for Operational Excellence.
- Developed EPANET software based digital twin of large scale water distribution networks using the differential evolution (DE) optimization based calibration technique.

---

- The DT used for application of Graph Convolution Net (GCN) based pipe leak localization and water quality regulation on case study of benchmark C-Town water network.
- Built framework for fault detection and diagnosis using combined Graph Auto-Encoder (GAE) and of Attention based Graph Neural Networks (GNN) for multivariate sensor systems.

### Gujarat State Fertilizer and Chemicals Ltd. (GSFC), India

2020–2021

Plant Engineer- Process

- Effectively coordinated with experienced professionals of the Technical Services team to delegate projects on optimization of flue gas ratio to waste heat recovery boiler and flare such that recollect more energy with minimum OPEX.

### Indian Oil Corporation Ltd. (IOCL)– Gujarat Refinery, India

May 2019–Jun 2019

Summer Intern

- Got on-site exposure to Crude Distillation unit, High-pressure reactors, Steam reformer, Heat exchangers, and PSA units. In-depth study of the process flow diagram, verified Mass Balance-Energy Balance of the Hydro cracker unit.

## PUBLICATIONS

- Tsai, ML., **Brahmbhatt, P.** and Avraamidou, S., 2026. Parametric Region Search: A Mixed-Integer Bilevel Optimization Problem Primal Heuristic. *arXiv*. [\[preprint\]](#)
- **Brahmbhatt, P.**, Ganesh, H.S. and Avraamidou, S., 2026. Explicit Distributed MPC: Reducing Computation and Communication Load by Exploiting Facet Properties. *arXiv*. [\[preprint\]](#)
- Ramesh, U.K., **Brahmbhatt, P.**, Quam, G.D., Avraamidou, S. and Ganesh, H.S., 2025. Optimal power dispatch from battery and engine of a hybrid vehicle through multiparametric mixed-integer programming. In 2025 Eleventh Indian Control Conference (ICC) (pp. 701-706). IEEE. [\[link\]](#)
- **Brahmbhatt, P.**, Cole, D.L., Zavala, V.M. and Avraamidou, S., 2025. Benders Decomposition using Graph Modeling and Multi-Parametric Programming. *Industrial & Engineering Chemistry Research*. [\[link\]](#)
- Vergara-Zambrano J., **Brahmbhatt P.**, Avraamidou S., 2025. A multi-scale optimization framework for energy transition planning in urban areas: Insights from a university campus case study. *Sustainable Energy, Grids and Networks*, p.101996. [\[link\]](#)
- Saini, R.S., **Brahmbhatt, P.**, Avraamidou, S. and Ganesh, H.S., 2025. Iteration-free cooperative distributed model predictive control through multiparametric programming. *Computers & Chemical Engineering*, p.109169. [\[link\]](#)
- **Brahmbhatt, P.**, Patel, R., Maheshwari, A. and Gudi, R.D., 2024. Improved Fault Detection and Diagnosis Using Graph Auto Encoder and Attention-based Graph Convolution Networks. *Digital Chemical Engineering*, p.100158. [\[link\]](#)
- **Brahmbhatt, P.**, Maheshwari, A. and Gudi, R.D., 2023. Digital twin assisted decision support system for quality regulation and leak localization task in large-scale water distribution networks. *Digital Chemical Engineering*, 9, p.100127. [\[link\]](#)

## CONFERENCE PRESENTATIONS

### Oral Presentations

- **Brahmbhatt, P.**, Cole, D.L., Zavala, V.M. and Avraamidou, S. Graph-Structured Optimization with Multi-Parametric Programming Surrogates for Benders Decomposition. In 2025 AIChE Annual Meeting. AIChE, Boston, MA.

- **Brahmbhatt, P.**, Ganesh, H.S., Avraamidou, S., Accelerating Distributed MPC Via Facet Properties: Faster Computation with Lower Communication Overhead. In 2025 AIChE Annual Meeting. AIChE, Boston, MA.
- **Brahmbhatt, P.**, Jennings, K., McCullough, A., Zavala, V. and Avraamidou, S., Multi-Market Bidding and Demand Response for Redox Reservoir-Based Electrochemical Devices Using Two-Stage Stochastic Optimization. In 2025 AIChE Annual Meeting. AIChE, Boston, MA.
- **Brahmbhatt, P.**, Cole, D.L., Zavala, V.M. and Avraamidou, S. Benders Decomposition using Graph Modeling and Multi-Parametric Programming. *Texas-Wisconsin-California Control Consortium, Fall 2025 Meeting*, Madison, WI.
- **Brahmbhatt, P.**, Saini, R.S., Ganesh, H.S., Avraamidou, S. Iteration-free cooperative distributed model predictive control through multiparametric programming. *Texas-Wisconsin-California Control Consortium, Fall 2024 Meeting*, Madison, WI.

## Poster Presentations

- Quam, G., Uthraa, K.R., **Brahmbhatt, P.**, Ganesh, H.S., Avraamidou, S. Optimal Power Dispatch of a Hybrid Vehicle through Multiparametric Mixed-Integer Programming *Texas-Wisconsin-California Control Consortium, Fall 2025 Meeting*, Madison, WI.
- **Brahmbhatt, P.**, Jennings, K., McCullough, A., Zavala, V. and Avraamidou, S., Multi-Market Bidding and Demand Response for Redox Reservoir-Based Electrochemical Devices Using Two-Stage Stochastic Optimization. Power Grid-Informed Electrochemistry Program 2025, Madison, WI.
- **Brahmbhatt, P.**, Quam, G., Uthraa, K.R., Ganesh, H.S., Avraamidou, S. ReLU Neural Network-based Explicit Model Predictive Control (MPC) *Data Science Research Bazaar 2025*, Madison, WI.
- **Brahmbhatt, P.**, Uthraa, K.R., Ganesh, H.S., Avraamidou, S. ReLU-Neural Network-based Explicit MPC: Case Study of Wastewater Treatment Plant. *Texas-Wisconsin-California Control Consortium, Fall 2024 Meeting*, Madison, WI.

## PROJECTS

### Bi-level Optimization Problem Generator

May 2025–July 2025

- Developed a versatile generator for bilevel optimization problems, supporting various combinations of LP/QP and MILP/MIQP at both the upper and lower levels.
- Validated problem non-triviality using high-point relaxation methods and solved for global solutions with multi-parametric programming.

### Performance Evaluation of PID, Model Predictive Controller (MPC), Pole Placement Controller (PPC), and Linear Quadratic Optimal Controller (LQOC) Using Nonlinear System Dynamic Simulation for the Reactor System

Jan 2022–May 2022

- Developed linear quadratic optimal controller (LQOC) using a data-driven state space ARX model and model predictive controller (MPC) using a linearized mechanistic model and Kalman filter in MATLAB.
- Servo and regulatory performances of the state feedback controllers and conventional multi-loop PID were compared by simulating the closed-loop dynamics using the nonlinear reactor model.

### Vehicle Traffic Prediction with Spatial-Temporal Graph Convolution Neural Networks using multivarite sensor data

Aug 2022–Nov 2022

- Studied fundamentals of Graph Convolutions and Graph Attention mechanism and reviewed recent research on DSTAGNN: Dynamic Spatial-Temporal Aware Graph Neural Network for Traffic Flow Forecasting.

- Fetched for METR-LA (Los Angeles Metropolitan Traffic) dataset in Pytorch Geometric Temporal Data sequential loader and developed ASTGCN Neural Net model; uses Spatial and Temporal aggregation as Graph Convolution (GNN) and Gated Recurrent Unit (GRU) respectively.

## TEACHING EXPERIENCE

- Teaching Assistant: Process Dynamics and Control, UW-Madison, Spring 2024 and Fall 2024.
- Teaching Assistant: Process Control (Spring 2023), Transport Phenomenon (Fall 2022), Process Equipment Design (Spring 2022); IIT- Bombay.

## AWARDS & SERVICE

- Officer team member at Chemical Engineering Graduate Student Association (ChEGS), *UW Madison*, 2025-2026.
- Bird, Stewart, and Lightfoot Graduate Fellowship Fund #2 for 1<sup>st</sup> year, *Chem. & Bio. Eng. Department, UW-Madison*, Fall 2023.
- GATE Scholarship by Ministry of Human Resource and Development, India, *IIT- Bombay*, 2021–2023.
- The M.S. University Gold Medal for exceptional academic performance in B.E. Chemical Engineering, *The Maharaja Sayajirao University of Baroda*, 2020.
- MYSY Scholarship, *The Maharaja Sayajirao University of Baroda*, 2016.
- Training and Placement coordinator, *MSU- Baroda*, 2019-2020.
- General Secretary, Association of Chemical Engineering Student, *MSU- Baroda*, 2017–2019.

## REFERENCES

- Prof. Styliani Avraamidou (PI)  
Assistant Professor, Chemical and Biological Engineering Department,  
University of Wisconsin-Madison.  
Email: [avraamidou@wisc.edu](mailto:avraamidou@wisc.edu)
  - Prof. Victor Zavala  
Professor, Chemical and Biological Engineering Department,  
University of Wisconsin-Madison.  
Email: [victor.zavala@wisc.edu](mailto:victor.zavala@wisc.edu)
  - Prof. Ravindra Gudi  
Professor, Department of Chemical Engineering,  
Indian Institute of Technology Bombay.  
Email: [ravigudi@che.iitb.ac.in](mailto:ravigudi@che.iitb.ac.in)
-